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Investigation of dual absorbers with novel MXene doped perovskite for Extraordinary performance of perovskite solar cells

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ABSTRACT

This study focuses on the numerical modeling of methylammonium lead halide ($MAPbl_{3\text{-}X}Cl_X$) based perovskite solar cells (PSCs) under optimal conditions. The selection of two perovskite materials, $MAPbl_{3\text{-}X}Cl_X$ and $MAPbI_3 + Ti_3C_2$, as the light absorber is advantageous due to their ability to achieve a broader absorption spectrum, with both materials having a bandgap in the range of 1.55 eV–1.6 eV, which is comparable to that of methylammonium lead iodide ($MAPbI_3$). While these materials still contain lead and thus share the same toxicity concerns, the introduction of MXenes like Ti_3C_2 enhances stability, improves charge transport, and increases overall efficiency, making them more viable for long-term applications. To further enhance device efficiency, selecting stable and high-performing carrier transport materials (CTMs) is a key strategy. Among the proposed options, the combination of Spiro-OMeTAD and ZnO as CTMs, along with an optimized thickness of the $MAPbl_{3\text{-}X}Cl_X$ and $MAPbI_3 + Ti_3C_2$ layers, resulted in a higher power conversion efficiency (PCE) of 27.12 % under AM1.5 photo illumination. Moreover, minimizing defects in PSC devices is crucial for further optimization and future advancements.

1. Introduction

Perovskite solar cells (PSCs) have garnered significant attention in renewable energy due to their high-power conversion efficiencies (PCEs) and cost-effective fabrication processes [1,2]. Since their inception, PSCs have rapidly surpassed traditional silicon-based solar cells in several performance metrics, thanks to the unique properties of perovskite materials, such as broad absorption spectra, high absorption coefficients, and favorable charge transport [3,4]. The ability to fine-tune the optoelectronic properties of these materials through compositional adjustments has further accelerated their development in photovoltaic technologies.

However, PSCs based on methylammonium lead iodide (MAPbI3) face critical challenges related to stability. While these lead-based perovskites offer impressive PCEs, they are prone to environmental degradation due to moisture, oxygen, and thermal stress [5–7]. This instability, coupled with the environmental and health risks posed by lead leakage, limits the practical deployment of MAPbI3-based PSCs. Addressing these issues is crucial for their commercialization and long-term viability as a sustainable energy solution.

Recent advancements aim to overcome these limitations by exploring alternative perovskite compositions and incorporating novel materials to enhance device stability and performance. For instance, the integration of MXenes—materials known for their excellent electrical conductivity and stability—into PSCs has shown promise in improving charge transport and mitigating degradation pathways [8–13]. These strategies, including modifying the perovskite's halide composition and adding protective layers, are essential for advancing PSC technology

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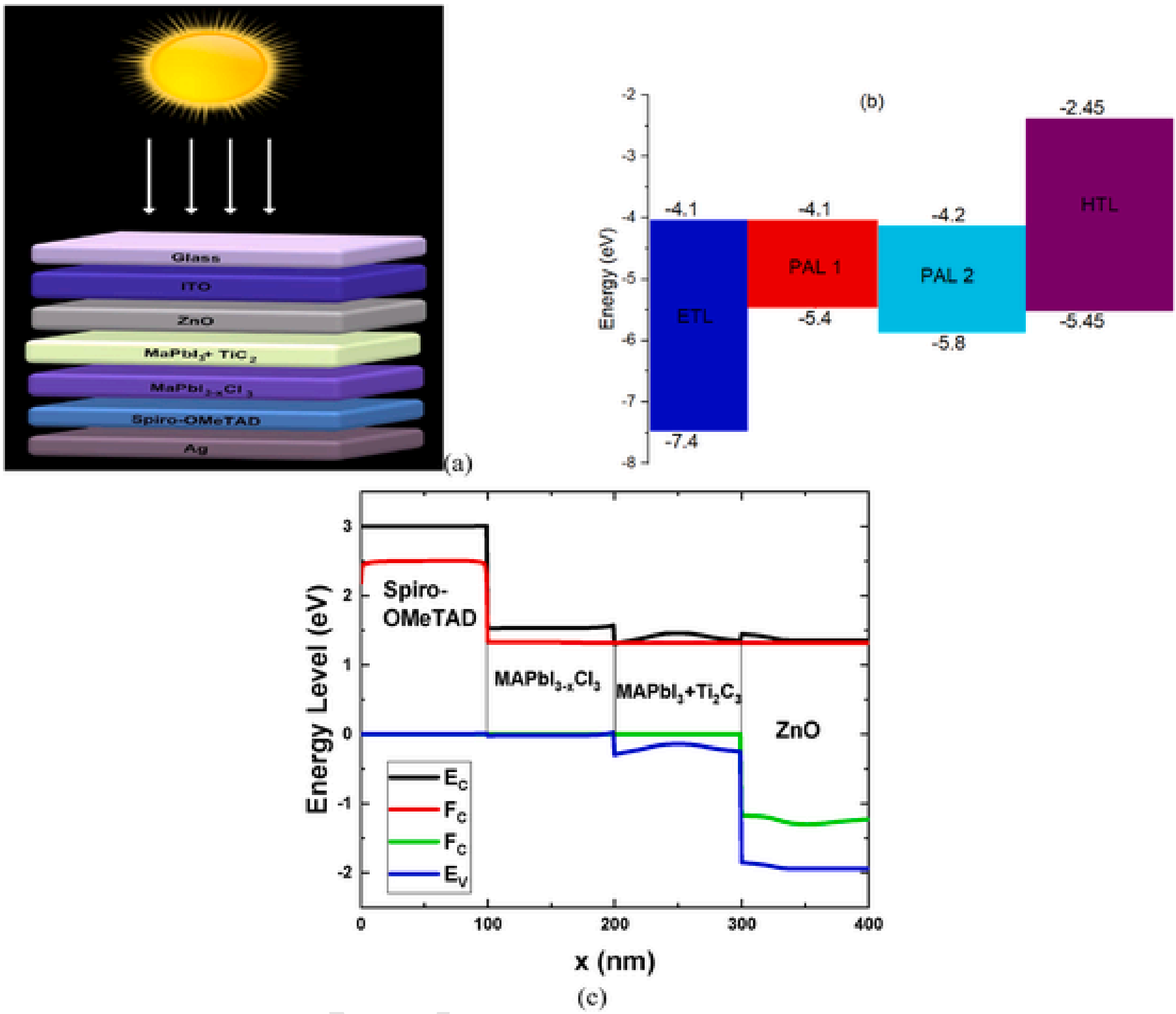

Fig. 1. (a) Representation of schematics layer of the device structures, (b) working mechanism along with the energy band of layers, and (c) band alignment of the device layers under the illuminance of AM1.5.

and ensuring its long-term sustainability in the global renewable energy landscape.

Recent advancements in perovskite materials have focused on overcoming the stability and performance limitations of traditional MAPbI3-based solar cells by exploring alternative compositions and incorporating novel materials [14–16]. One promising approach is the use of methylammonium lead halide (MAPbl3-XClX) as an alternative perovskite layer. With a tunable bandgap between 1.55 eV and 1.6 eV, MAPbl3-XClX offers a broader absorption spectrum compared to MAPbI3 and improved stability due to the incorporation of chlorine, which enhances moisture resistance and reduces degradation rates [17,18].

The integration of Ti3C2 MXenes into the perovskite layer has shown significant potential in enhancing the overall performance of solar cells [19,20]. Ti3C2 MXenes, known for their high electrical conductivity and stability, improve charge transport and reduce recombination losses, addressing key performance and durability issues in PSCs [20]. The combination of MAPbI3 and Ti3C2 MXenes represents a strategic advancement in developing high-efficiency and stable PSCs. Leveraging the broad absorption range of MAPbI3 and the superior electronic properties of Ti3C2 MXenes can lead to higher PCE and improved device longevity [21–23]. This integration not only mitigates the degradation issues of traditional perovskite materials but also optimizes charge transport and reduces defect-related losses.

Ti3C2 MXenes have emerged as a significant advancement in optimizing PSCs due to their unique combination of electronic and structural properties. Their exceptional electrical conductivity and high surface area facilitate superior charge transport within the solar cell. The 2D structure of Ti3C2 MXenes enhances carrier mobility by providing an efficient pathway for electron and hole movement, reducing recombination losses and improving the overall PCE of the device [19]. Acting as a robust interfacial layer, Ti3C2 MXenes improve energy level alignment between the perovskite layer and carrier transport materials (CTMs), optimizing charge extraction and minimizing energy losses. Additionally, the chemical stability and resistance of MXenes to environmental degradation help mitigate the inherent instability of traditional perovskite materials. By serving as a protective barrier, Ti3C2 MXenes shield the perovskite layer from moisture, oxygen, and thermal stress, extending the operational lifetime of solar cells. This enhanced stability, combined with improved charge transport properties, makes Ti3C2 MXenes a valuable component in advancing PSC technology, potentially leading to more durable and efficient solar energy solutions [21,22].

Optimizing CTMs is crucial for enhancing PSC performance by improving charge extraction and minimizing recombination losses. CTMs facilitate efficient charge collection from the perovskite layer to the electrodes. Spiro-OMeTAD, a widely used hole transport material (HTM), is renowned for its high hole mobility and excellent film-

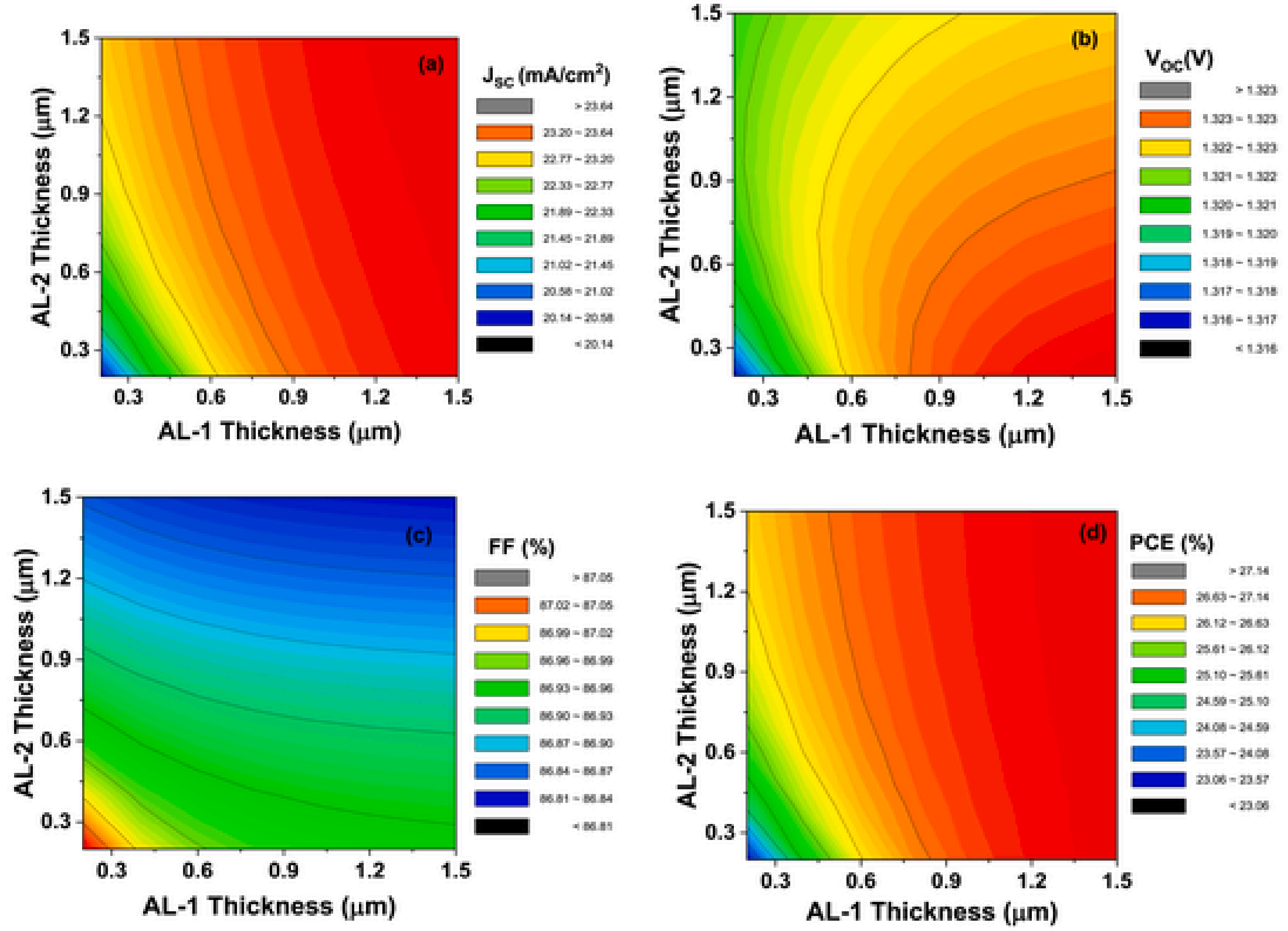

Fig. 2. Variation of electrical parameters, namely (a) short circuit current density (J_{SC}), (b) open-circuit voltage (V_{OC}), fill factor (FF), and (d) power conversion efficiency (PCE) over the function of PAL 1 and PAL 2 thickness of the PSC.

Table 1
THE SIMULATION PARAMETERS AND MATERIAL PROPERTIES IN MAPbI3-XClx and MAPbI3 + Ti3C2BASED PSC DEVICE [28–30].

Parameters	ZnO	$MAPbI_{3-x}Cl_x$	$MAPbI_3$ + Ti_3C_2	Spiro-OMeTAD
t	0.1–0.4	0.3–1.5	0.3–1.5	0.1–0.4
E_g (eV)	3.3	1.3	1.6	3.0
χ (eV)	4.1	4.1	4.2	2.45
ε_r	9	8.2	6.5	3.0
N_c (cm^{-3})	4×10^{18}	1×10^{18}	1×10^{19}	1×10^{19}
N_v (cm^{-3})	1×10^{19}	1×10^{18}	1×10^{19}	1×10^{19}
μ_n (cm^2/Vs)	100	1.6	2	2×10^{-4}
μ_p (cm^2/Vs)	50	1.6	2	2×10^{-4}
N_d (cm^{-3})	1×10^{18}	0	0	0
N_a (cm^{-3})	1×10^{5}	1×10^{15}	7×10^{15}	2×10^{18}
N_t (cm^{-3})	1×10^{14}	1×10^{14}–1×10^{15}	1×10^{14}–1×10^{15}	1×10^{14}

forming properties, ensuring effective hole extraction and transport in PSCs [24,25]. Its ability to form a uniform, pinhole-free layer enhances the interface between the perovskite absorber and the electrode, reducing charge recombination and improving overall device efficiency. Similarly, zinc oxide (ZnO) serves as an effective electron transport material (ETM) due to its high electron mobility and stability. ZnO forms a conductive and stable interface with the perovskite layer, facilitating efficient electron extraction while minimizing energy losses. The choice

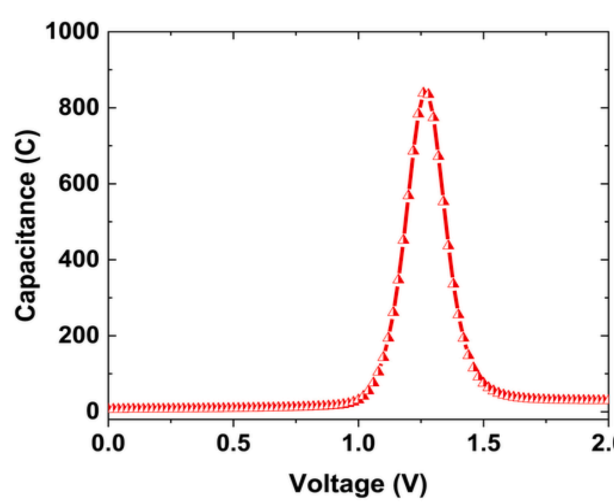

Fig. 3. C–V curve of the modeled device.

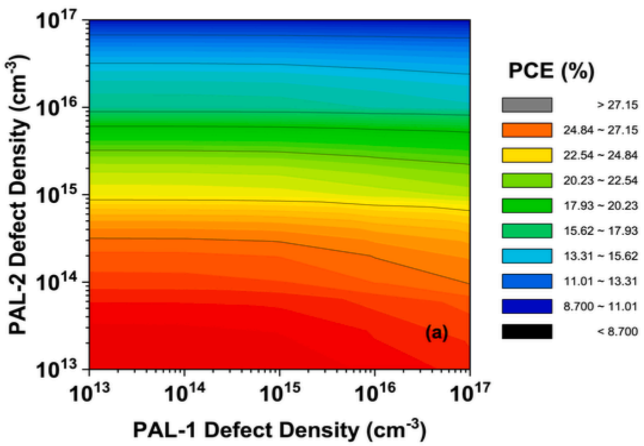

Fig. 4. Effect of defect density of PAL 1 and PAL 2 on the PCE of the device.

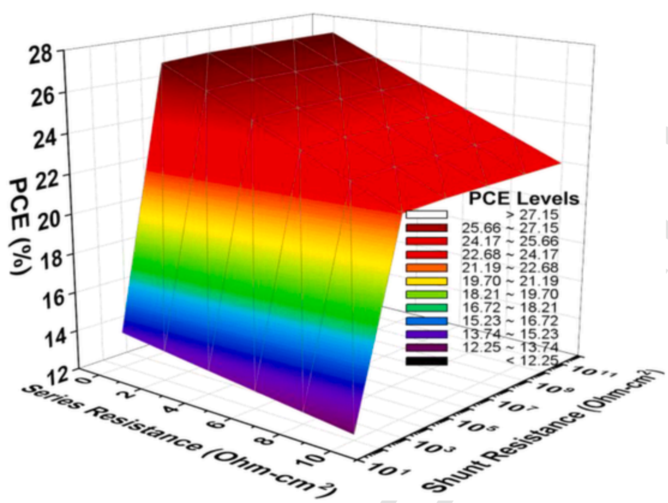

Fig. 5. Variation of PCE as a function of series and shunt resistance of the device.

of ZnO as an ETM is advantageous due to its low cost, ease of processing, and tunability for optimal performance [26,27]. Together, Spiro-OMeTAD and ZnO provide a complementary set of properties that optimize charge transport in PSCs, leading to enhanced PCEs and improved device stability.

The objective of this study is to advance PSC performance through comprehensive numerical modeling and optimization of device parameters. By investigating different perovskite materials, specifically MAPbI3-XClX and MAPbI3 + Ti3C2, and integrating high-performance CTMs like Spiro-OMeTAD and ZnO, the research aims to enhance PCE and stability in PSCs. Advanced modeling techniques simulate the electrical and optical properties of these materials, optimize layer thicknesses, and evaluate overall device performance under various conditions. The goal is to identify the most effective material combinations and configurations to achieve superior efficiency and durability, contributing to the advancement of perovskite solar technology.

This article is structured to provide a comprehensive analysis of PSC optimization through numerical modeling and material selection. The first section introduces the theoretical framework and background of PSC technology, highlighting the challenges of traditional perovskite materials and the potential benefits of alternative compositions. Subsequent sections detail the methodologies used for modeling and simulation, including the selection of perovskite materials such as MAPbI3-XClX and MAPbI3 + Ti3C2, and the integration of CTMs like Spiro-OMeTAD and ZnO. The results section presents findings on device performance, including PCEs and stability assessments, followed by a discussion of how these findings contribute to advancements in PSC technology.

2. Device configuration of the PSC

The PSC device architecture in this study is meticulously designed to optimize both efficiency and stability (Fig. 1a). The structure starts with a transparent conductive oxide (TCO) layer, typically indium tin oxide (ITO), serving as the front electrode, which allows light to enter the cell. Next is the electron transport layer (ETL), made of zinc oxide (ZnO), selected for its high electron mobility and chemical stability. The ZnO layer efficiently extracts electrons from the perovskite absorber layer and channels them toward the electrode, minimizing recombination losses (see Fig. 2).

The core of the device features a dual perovskite absorber layer, comprising $MAPbI_{3\text{-}X}Cl_X$ (PAL 1) and $MAPbI_3 + Ti_3C_2$ MXene (PAL 2). This combination is chosen to achieve a broad absorption spectrum and enhance charge carrier dynamics. $MAPbI_{3\text{-}X}Cl_X$ captures a wide range of the solar spectrum, while the incorporation of Ti_3C_2 MXenes into $MAPbI_3$ improves charge transport, increases carrier mobility, and enhances the overall stability of the perovskite layer.

The absorber layer is sandwiched between the ZnO ETL and the hole transport layer (HTL), Spiro-OMeTAD, known for its high hole mobility and excellent film-forming properties. Spiro-OMeTAD facilitates effective hole extraction from the perovskite and transport to the back electrode. The architecture is completed with a back electrode, typically gold or silver, which collects the holes and closes the electrical circuit.

The integration of Ti_3C_2 MXenes within the perovskite structure plays a crucial role in enhancing the PCE and improving the long-term operational stability of the PSC. This advanced architecture represents a significant step forward in developing highly efficient and stable solar energy technologies (Fig. 1b and c, Table 1). These parameters have been carefully selected from previously published research articles to ensure accuracy and reliability in the modeling process. The choice of accurate input parameters is critical in a modeling study as it directly influences the simulation outcomes, which in turn guide the design and optimization of the actual device. By drawing on established values from prior studies, the modeling ensures that the simulated conditions closely resemble real-world scenarios, leading to more precise predictions of device performance. Accurate input parameters, such as the bandgap, electron affinity, and carrier mobility, are essential for understanding the interactions within the PSC, optimizing the layer thicknesses, and ultimately enhancing the power conversion efficiency (PCE) and stability of the device. This approach not only builds on the foundation of existing research but also ensures that the developed models are robust and capable of guiding future experimental work in PSC development. The study utilizes the AM 1.5G solar spectrum for device modeling, maintaining a temperature of 300K. The absorption spectra of the PAL 1 and PAL 2 materials were generated using the SCAPS 1D software, incorporating several approximations.

3. Results & discussion

A. Impact of PAL thickness on device performance

The impact of perovskite absorber layer (PAL) thickness on the performance of PSCs has been systematically evaluated through a contour plot analysis, focusing on key performance metrics: short-circuit current density (J_{SC}), open-circuit voltage (V_{OC}), fill factor (FF), and power conversion efficiency (PCE). The analysis examined the thickness of PAL 1 ($MAPbI_{3\text{-}X}Cl_X$) on the X-axis and PAL 2 ($MAPbI_3 + Ti_3C_2$ MXene)

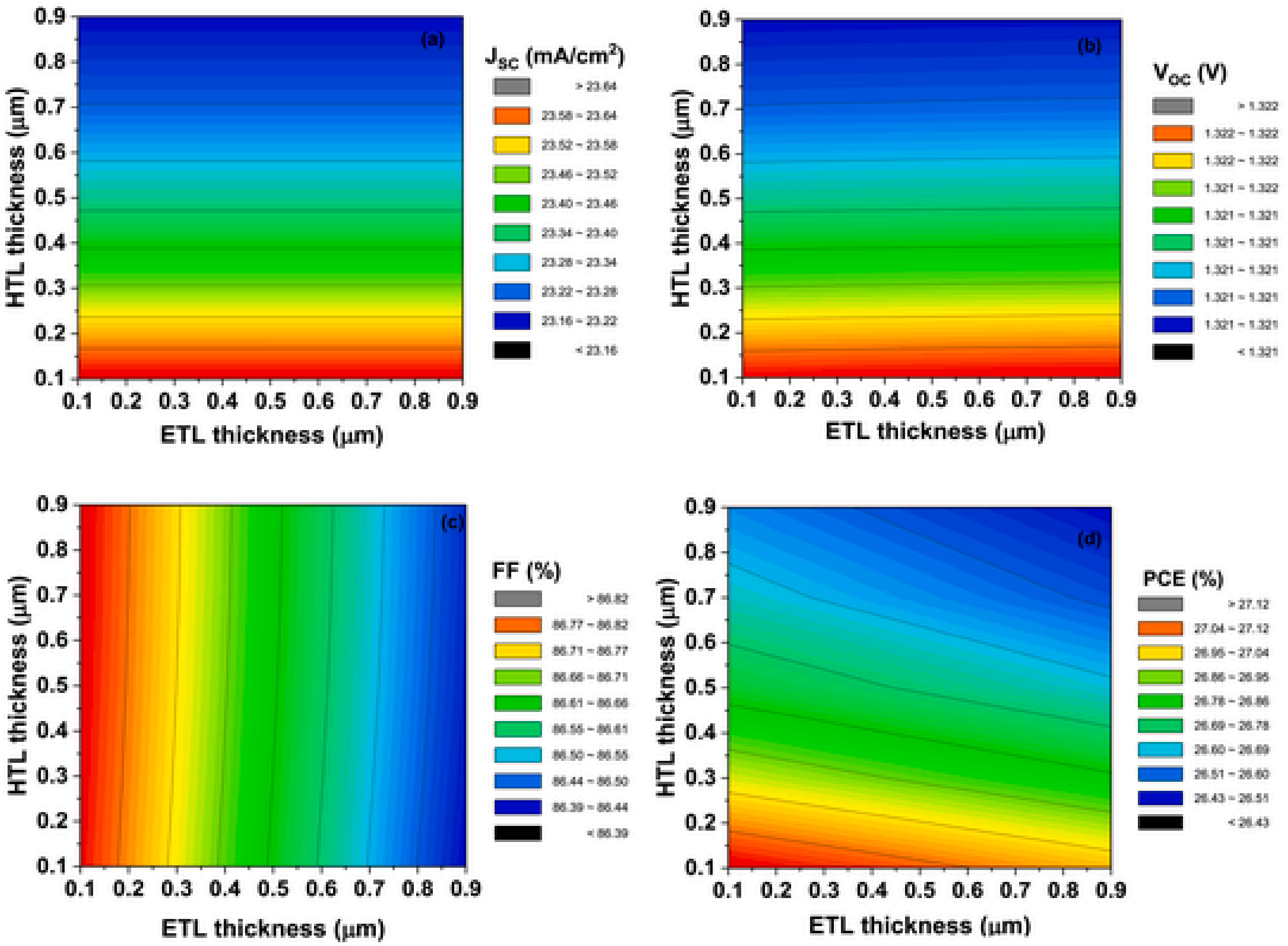

Fig. 6. The PSC characteristics of (a) J_{SC}, (b) V_{OC}, (c) FF, and (d) PCE with the thickness variation of ETL and HTL.

on the Y-axis, revealing how variations in these layers affect device performance. For J_{SC}, it was found that optimal values are achieved when PAL 1 is thicker, ranging from 1.2 μm to 1.5 μm, while PAL 2 remains thinner, between 0.25 μm and 1.5 μm. This suggests that a thicker PAL 1 enhances light absorption and charge collection, while a thinner PAL 2 improves charge transport and reduces recombination losses. This balance increases the J_{SC}, making the dual absorber layer more effective. In contrast, the V_{OC} is maximized when PAL 1 is also thicker (1.2–1.5 μm), but PAL 2 is thinner (0.25–0.4 μm). This combination provides an optimal balance between light absorption and charge carrier extraction, resulting in a strong built-in electric field and reduced carrier recombination at the interface. These factors contribute to a higher V_{OC}. Regarding the fill factor (FF), the study found that thinner layers for both PAL 1 and PAL 2 result in optimal FF values. Thinner absorber layers reduce series resistance and enhance charge transport efficiency, allowing charge carriers to move through the device with less resistance. This improves overall charge extraction efficiency and minimizes bulk recombination, leading to a higher FF. In summary, the optimal performance of PSCs is achieved by carefully adjusting the thickness of PAL 1 and PAL 2, with thicker PAL 1 and thinner PAL 2 being favorable for J_{SC} and V_{OC}, while thinner layers for both are beneficial for FF. These findings provide valuable guidance for optimizing layer thickness in PSCs to enhance their efficiency and stability. The power conversion efficiency (PCE) of perovskite solar cells (PSCs) is a critical performance metric that closely correlates with short-circuit current density (J_{SC}). The study shows that PCE peaks when the thickness of PAL 1 is between 1.2 and 1.5 μm, while the thickness of PAL 2 can vary from 0.25 to 1.5 μm without significantly impacting the performance. This indicates that a thicker PAL 1 is essential for effective photon absorption and charge carrier generation, driving higher J_{SC} and, consequently, higher PCE. PAL 2's flexibility in thickness highlights its supportive role in charge extraction and recombination minimization, allowing the device to maintain high PCE across various configurations.

For optimal device performance, the study suggests that the combination of a thicker PAL 1 and a thinner PAL 2 is the most effective configuration. Specifically, a thicker PAL 1 in the range of 1.2–1.5 μm enhances light absorption and charge carrier generation, which are critical for improving both J_{SC} and PCE. Simultaneously, a thinner PAL 2, ranging from 0.25 to 0.4 μm, facilitates efficient charge extraction and minimizes recombination losses, optimizing the fill factor (FF) and open-circuit voltage (VOC). This balanced design approach ensures that the solar cell achieves high performance, making it a promising strategy for future device designs aimed at maximizing efficiency and stability.

The capacitance-voltage (C–V) curve is another important diagnostic tool for understanding the electrical properties of PSCs, particularly the behavior of the p-n junction and the depletion region. The C–V curve for the modeled device shows peak capacitance values between 1 and 1.5 V, with capacitance dropping to zero at other voltage levels. This suggests a significant response of the depletion region within this voltage range, likely corresponding to the built-in potential of the solar

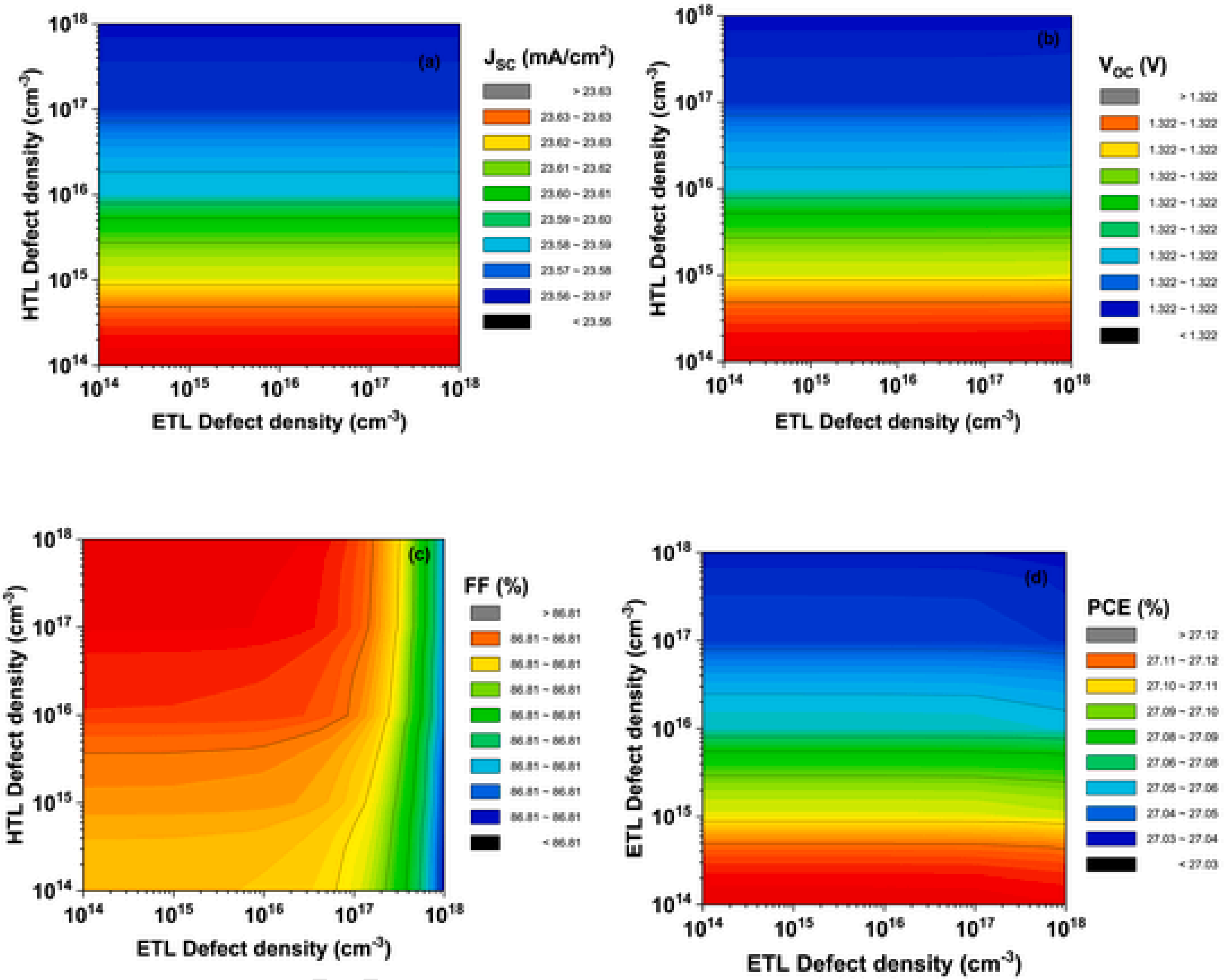

Fig. 7. Effect of defect density of ETM and HTM on the performance of the dual absorber PSC.

cell or the onset of strong carrier accumulation or depletion at the interface.

The observation that capacitance is zero outside this 1–1.5 V range implies that the device may be fully depleted or lacks significant charge separation at other voltages. This behavior indicates that the junction is only active or responsive within this specific voltage window, which may relate to the unique properties of the absorber materials used in the device. It also suggests that the effective operational voltage range for charge separation is narrow, pointing to a potential area for further optimization to broaden the effective operational range of the solar cell. This insight is crucial for future enhancements, aiming to improve the efficiency and robustness of PSCs across a wider voltage range.

B. Impact of PAL defect density on device performance

Defect density in the absorber layers of a solar cell is a critical parameter that significantly influences device performance (see Fig. 3). A lower defect density indicates fewer imperfections and recombination centers within the absorber material, which directly enhances charge carrier dynamics. Defects such as traps and recombination centers can capture and recombine charge carriers, leading to energy losses and reduced photocurrent. By minimizing defect density, the solar cell achieves improved charge carrier mobility and collection efficiency, resulting in higher J_{SC} and V_{OC}. A lower defect density reduces series resistance and improves FF, leading to a higher overall PCE. Therefore, optimizing defect density is essential for maximizing the efficiency and stability of PSCs, ensuring effective energy conversion and long-term performance. The study indicates that the device achieves its maximum PCE of 27.12 % when the defect densities of PAL 1 and PAL 2 are in the lower range (10^{13} cm^{-3} to 10^{15} cm^{-3}) as depicted in Fig. 4. This outcome is primarily due to the significant impact of defect density on charge carrier dynamics and overall device performance. Lower defect densities in the absorber layers reduce the number of recombination centers that can trap and recombine charge carriers. This leads to fewer losses in generated charge carriers, thereby enhancing both the J_{SC} and V_{OC}. Fewer defects contribute to a higher FF by minimizing resistance losses and improving charge collection efficiency. Consequently, with lower defect densities, the solar cell can operate more efficiently, translating into a higher PCE.

In PSCs, series resistance (R_S) and shunt resistance (R_{Sh}) are critical parameters that significantly influence device performance. R_S is associated with the resistive losses in the cell's components, including the electrodes, the transport layers, and the perovskite absorber material itself. High R_S leads to a reduction in the FF and PCE by impeding the

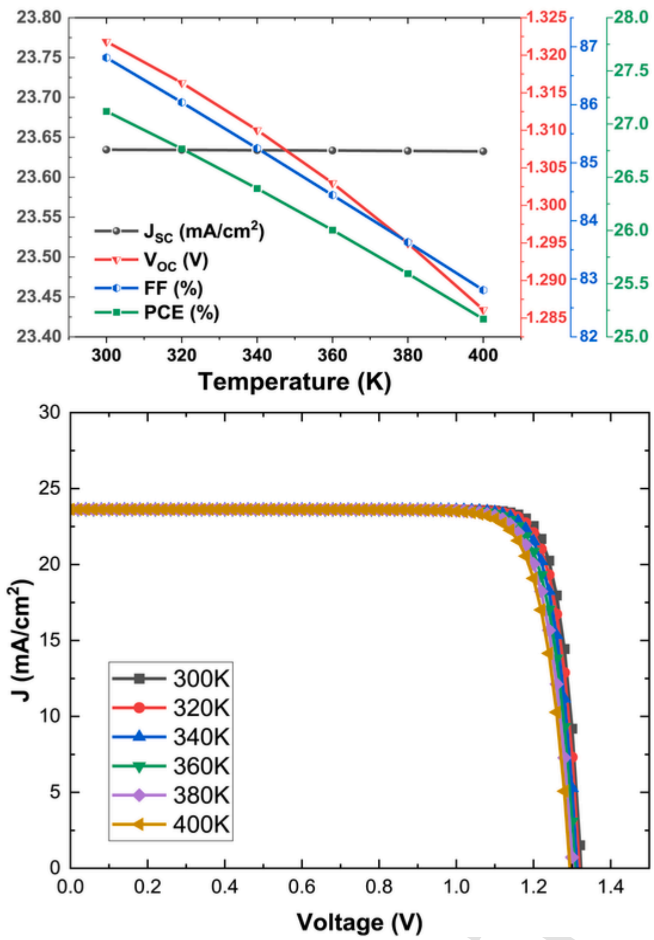

Fig. 8. PV parameters and JV parameters of the device as a function of temperature.

flow of carriers, which can result from defects in the absorber layer. These defects create barriers to charge transport, increasing resistance within the device. On the other hand, R_{Sh} refers to the parallel pathways within the solar cell where unwanted current can bypass the active regions, usually due to defects or imperfections in the absorber material or at the interfaces. Low R_{Sh} causes leakage currents, which reduce the V_{OC} and FF, thereby lowering the PCE of the PSC. High defect density in the absorber material can increase the number of pathways for shunt currents, leading to a decrease in R_{Sh}. Thus, reducing defect density in the absorber layer is crucial for minimizing R_S by improving carrier mobility and for maximizing R_{Sh} by reducing leakage currents. This optimization leads to enhanced FF, V_{OC}, and overall PCE, resulting in more efficient PSCs. Fig. 5 depicts the variation of PCE of the device as a function of R_S and R_{Sh} of the device.

C. Effect of thickness and defect density of ETL and HTL

Both ETL and HTL are responsible for selectively transporting electrons and holes, respectively, while blocking the opposite charge carriers, which helps in reducing recombination losses. The thickness of the ETL impacts the charge extraction efficiency and the overall resistance in the device. An optimal ETL thickness ensures efficient electron collection and transport from the perovskite absorber to the electrode. If the ETL is too thin, it may not fully cover the perovskite layer, leading to incomplete charge extraction and increased recombination. Conversely, if the ETL is too thick, it can introduce additional series resistance, impeding electron transport and reducing the FF and PCE. Similarly, the HTL's thickness affects the transport and collection of holes. An optimally thick HTL provides a sufficient pathway for holes to reach the electrode while blocking electrons. A thin HTL might not adequately cover the perovskite layer, leading to poor hole extraction and increased recombination at the interface. On the other hand, a very thick HTL can increase the series resistance, which again reduces the FF and PCE. An overly thick HTL may also lead to light absorption losses, reducing the overall photocurrent.

In this study, the thicknesses of the ETL and HTL were varied between 0.1 and 0.9 μm to assess their impact on the key performance parameters of the PSC (Fig. 6). The results indicate that the J_{SC} remains constant across the entire range of ETL thicknesses. This stability in J_{SC} suggests that within the given thickness range, the ETL efficiently facilitates electron transport without introducing significant resistive losses or recombination at the interface, thus ensuring consistent electron extraction from the perovskite absorber layer. In contrast, the J_{SC} decreases with increasing HTL thickness beyond 0.1 μm. This reduction in J_{SC} can be attributed to the increased resistive losses and potential recombination sites that arise as the HTL becomes thicker. A thicker HTL

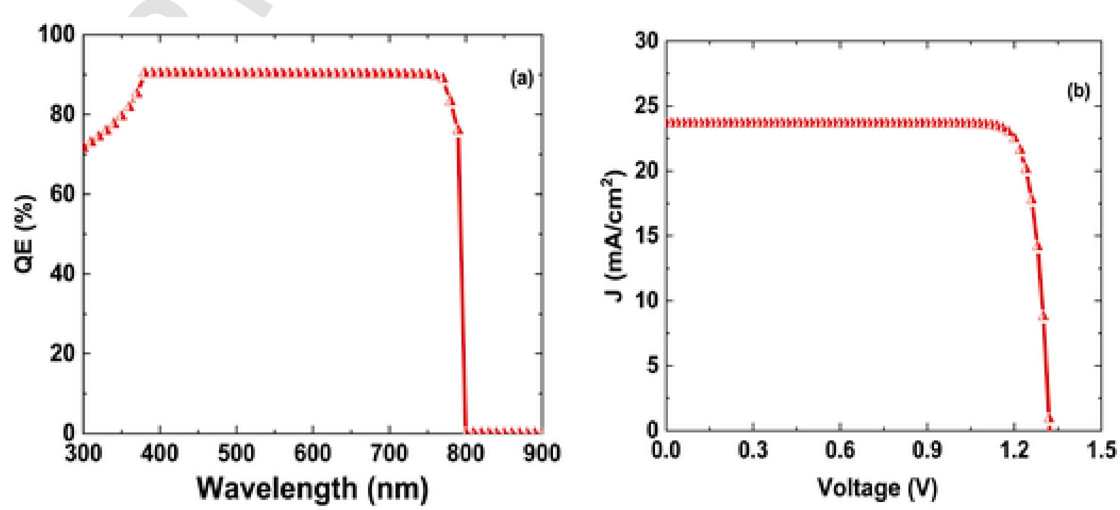

Fig. 9. (a) QE and (b) J-V characteristics of the dual absorber PSC.

Table 2
The comparison of different simulation results.

Device Structure	J_{SC} (mA/cm^2)	V_{OC} (V)	FF (%)	PCE (%)
$ZnO/MAPbI_{3-x}Cl_x$/Spiro-OMeTAD	16.92	1.37	88.63	20.62
$ZnO/MAPbI_{3-x}Cl_x/MAPbI_3+Ti_3C_2$/Spiro-OMeTAD	23.63	1.32	86.81	27.12
$ZnO/MAPbI_{3-x}Cl_x/MAPbI_3$/Spiro-OMeTAD (Simulated)	31.17	1.1	60.30	20.86
$ITO/C_{60}/MASnI_3/RbGeI_3/Cu_2O/Au$ [31]	32.7	0.914	65.49	20.19
$Glass/FTO/TiO2/MAPbI_{3-x}Cl_x$/Spiro-OMeTAD/BC [32]	21.20	1.02	77	16.7
$FTO/CdS/CH_3NH_3PbI_3$/Spiro-OMeTAD/Au [33]	24.35	1.37	71.29	23.83

can hinder the efficient transport of holes, resulting in lower charge collection efficiency.

In this study, as the thickness of the HTL increases from 0.1 μm, both V_{OC} and J_{SC} show a declining trend, while the FF exhibits an opposite behavior, increasing with HTL thickness. The reduction in V_{OC} with increasing HTL thickness can be explained by the fact that a thicker HTL introduces more resistive losses and recombination sites, which can lead to a higher rate of charge carrier recombination before they can be collected, thus reducing the overall voltage generated by the cell. As the HTL thickness increases, it may block the better transfer of the holes to the electrode. The PCE of the PSC reaches its optimal value when the ETL thickness is within the range of 0.1–0.4 μm, and the HTL thickness is between 0.1 and 0.2 μm. A thinner ETL in the specified range ensures that electrons generated in the absorber layer can be swiftly transported to the electrode, minimizing resistive losses and maintaining high electron mobility. However, if the ETL becomes too thick, it can introduce additional resistance and trap states, which would hinder electron transport and reduce efficiency. The optimal HTL thickness range ensures effective hole transport while minimizing charge recombination and resistive losses. When the HTL is too thick, the increased distance that holes must travel can lead to greater recombination losses and higher series resistance, negatively impacting PCE. Conversely, if the HTL is too thin, it may not adequately block electrons from reaching the anode, leading to leakage currents and further efficiency losses. Therefore, the identified thickness ranges for both ETL and HTL represent an ideal balance, facilitating maximum charge carrier extraction and minimal losses, resulting in the highest possible PCE for the device. The study reveals that the photovoltaic parameters V_{OC}, J_{SC}, and PCE remain unaffected by variations in the defect density of ETL within the range of 10^{14} cm^{-3} to 10^{18} cm^{-3}, suggesting that the ETL's performance is robust against defect-related losses in this range (Fig. 7). However, the same parameters show a decline when the defect density of HTL exceeds 10^{15} cm^{-3}, indicating that higher defect densities in the HTL lead to increased recombination losses, which negatively impact charge extraction and overall device efficiency. This trend highlights the critical role of maintaining low defect densities in the HTL to preserve high V_{OC}, J_{SC}, and PCE values. Interestingly, the FF shows an optimal value when the ETL has a lower defect density, which minimizes resistive losses, and when the HTL has a higher defect density. The higher defect density in the HTL might introduce some beneficial effects, such as improved charge blocking characteristics, which can help reduce leakage currents and optimize the FF. This complex interplay between defect densities and device performance parameters underscores the importance of carefully balancing the material properties of both ETL and HTL to achieve optimal performance in PSCs.

D. Effect of temperature on PSC devices

In the dual absorber PSC study, the performance parameters were analyzed by varying the device temperature from 300K to 400K. The results indicate that V_{OC}, FF, and PCE show a noticeable decline with increasing temperature, while the J_{SC} remains relatively constant with minimal deviation from its value at room temperature as depicted in Fig. 8. The decline in V_{OC} with rising temperature is primarily due to the increased rate of carrier recombination, which reduces the voltage generated by the cell. Similarly, the FF decreases because higher temperatures lead to an increase in R_S and a decrease in R_{Sh}, both of which negatively impact the device's ability to efficiently extract and utilize charge carriers. This, in turn, results in a reduced overall power PCE. The relative stability of J_{SC} across the temperature range suggests that the generation of photocurrent is not significantly affected by temperature, as it is more dependent on the absorption properties of the perovskite layers and less on the thermal behavior of the device. The combined effects emphasize the importance of thermal management in optimizing the performance and stability of PSCs. The modeled dual absorber PSC demonstrates impressive quantum efficiency (QE) of approximately 90 % across a broad wavelength range from 400 nm to 780 nm (Fig. 9a). This high QE indicates that the device is highly efficient at converting incident photons into electrical charge carriers within this spectral range. The efficient absorption and charge collection are likely due to the optimal bandgaps of the absorber materials, which are well-matched to the solar spectrum, allowing for maximum photon absorption in the visible region. The high QE across this range suggests that the solar cell is effectively utilizing the incident light, resulting in strong photocurrent generation, which is crucial for achieving high overall device efficiency. The broad spectral response also highlights the device's potential for effective energy conversion in a variety of lighting conditions, contributing to its high-performance metrics. Fig. 9b represents the J-V characteristics of the modeled device with dual perovskite absorber materials. Further, the comparison of different simulation results with the already published literature [29,30] has been reported in Table 2.

4. Conclusions

In conclusion, the current study demonstrates the potential of using $MAPbl_{3-X}CI_X$ and $MAPbI_3+Ti_3C_2$ as the absorber materials in PSCs to achieve enhanced performance. The broad absorption spectrum of these materials, coupled with the stability and charge transport improvements offered by the integration of Ti_3C_2 (MXenes), positions them as promising candidates for efficient and long-lasting PSCs. The careful selection and optimization of carrier transport materials, particularly the combination of Spiro-OMeTAD and ZnO, played a pivotal role in reaching a significant PCE of 27.12 %. The importance of minimizing defect densities to further boost device efficiency was underscored, highlighting the need for continued research into defect management and material optimization. This study provides valuable insights into the design and optimization of high-efficiency PSCs, leading to future advancements in photovoltaic technologies.

CRediT authorship contribution statement

Sagar Bhattarai: Writing – original draft, Formal analysis, Data curation. **K. Deepthi Jayan:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Prakash Kanjariya:** Writing – review & editing. **Pawan Sharma:** Writing – review & editing. **Ramneet Kaur:** Writing – review & editing. **Jaya Madan:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Mohd Zahid Ansari:** Funding acquisition, Writing – review & editing. **Saikh Mohammad Wabaidur:** Funding acquisition, Writing – review & editing. **Rahul Pandey:** Writing – review & editing, Visualization, Validation, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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